

Exercise and worked solution

Probability & Bayes' rule

Four parts to anchor the probabilistic foundations of the risk track: basic rules, Bayesian updating, expectation of a function, and how moments behave over time. Work through each part before reading the solution.

Part A · Basic rules

PROBLEM

Given $\Pr(A) = 0.3$, $\Pr(B) = 0.5$ and $\Pr(A \cap B) = 0.2$. Compute (a) $\Pr(A \cup B)$, (b) $\Pr(A | B)$, and (c) state whether A and B are independent.

SOLUTION

1. (a) Union rule: $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B) = 0.3 + 0.5 - 0.2 = \mathbf{0.6}$.
2. (b) Conditional: $\Pr(A | B) = \Pr(A \cap B) / \Pr(B) = 0.2 / 0.5 = \mathbf{0.4}$.
3. (c) Independence requires $\Pr(A \cap B) = \Pr(A) \cdot \Pr(B) = 0.3 \times 0.5 = 0.15$. The intersection is $0.2 \neq 0.15$, so A and B are **dependent** (positively, since $0.2 > 0.15$).

Part B · Bayes and the base-rate trap

PROBLEM

A default hits 2% of borrowers. An alert model catches 90% of true defaults (sensitivity) but also fires on 8% of healthy borrowers (false alarms). An alert fires. What is the probability the borrower truly defaults?

SOLUTION

1. True positives: $\Pr(\text{default}) \cdot \Pr(\text{alarm} | \text{default}) = 0.02 \times 0.90 = 0.018$.
2. False positives: $\Pr(\text{healthy}) \cdot \Pr(\text{alarm} | \text{healthy}) = 0.98 \times 0.08 = 0.0784$.
3. $\Pr(\text{alarm}) = 0.018 + 0.0784 = 0.0964$.
4. Bayes: $\Pr(\text{default} | \text{alarm}) = 0.018 / 0.0964 \approx \mathbf{18.7\%}$.
5. Reading: despite an alert, fewer than one in five is a true default. When the danger is rare, false alarms (from the many healthy) swamp the true ones. This is the **base-rate trap**, and why a sensitive signal is not enough without accounting for the base rate.

Part C · Expectation of a function and Jensen

PROBLEM

An asset's future price will be 20, 50 or 100 with probabilities 0.2, 0.5 and 0.3. (a) Compute the expected payoff of a call with strike 40, paying $\max(S - 40, 0)$. (b) Check the direction of Jensen's inequality on this example.

SOLUTION

1. (a) Payoffs per scenario: $\max(20-40, 0) = 0$; $\max(50-40, 0) = 10$; $\max(100-40, 0) = 60$. Expectation: $E[\text{payoff}] = 0.2 \times 0 + 0.5 \times 10 + 0.3 \times 60 = 0 + 5 + 18 = \mathbf{23}$.
2. (b) Expected underlying price: $E[S] = 0.2 \times 20 + 0.5 \times 50 + 0.3 \times 100 = 4 + 25 + 30 = 59$. Payoff of the expectation: $\max(59 - 40, 0) = 19$. We have $E[\text{payoff}] = 23 > 19 = \text{payoff}(E[S])$: consistent with Jensen, since a call payoff is **convex**. The expected payoff exceeds the payoff of the expectation: this is the value of optionality.

Part D · Moments and the time scale

PROBLEM

A monthly return has a mean of 1% and a standard deviation of 4%. (a) Annualise the mean and standard deviation (mean $\times 12$, standard deviation $\times \sqrt{12}$). (b) Explain why volatility scales in $\sqrt{t}$ and not t .

SOLUTION

1. (a) Annualised mean: $1\% \times 12 = \mathbf{12\%}$. Annualised standard deviation: $4\% \times \sqrt{12} \approx 4\% \times 3.464 = \mathbf{13.86\%}$.
2. (b) For independent returns, **variances** add over time: $\text{Var}(12 \text{ months}) = 12 \times \text{Var}(1 \text{ month})$. Since the standard deviation is the square root of the variance, it grows in $\sqrt{12}$, not 12. This is the rule $\text{Var}(a + bX) = b^2 \cdot \text{Var}(X)$: dispersion follows the square of the scale. Consequence: the return/risk ratio improves with horizon (numerator scales in t , denominator in $\sqrt{t}$).

Key takeaways: three axioms generate all the rules; Bayes updates a belief and exposes the base-rate trap; the expectation of a convex function exceeds the function of the expectation (Jensen); and volatility scales in the square root of time.